

The Price of Realism: A Filtration of the State Space and Three Studies of Its Transitions

Patrick S. Mahon*

Abstract

Finitely additive coherence is what observation natively supplies; σ -additivity and value-definiteness are further *representation properties* — signatures of a world, not of the data. The shared question is whether coherence, followed to its end, forces such signatures rather than assuming them, and, where it does, what exactly it forces. This umbrella note reads three companion papers as partial answers to that question, organised as one classification of the price of those signatures. The organising object is a filtration of the state space, $\mathcal{S} \supseteq \mathcal{S}_\sigma \supseteq \mathcal{S}_{\text{df}}^\sigma$, whose three transitions — *pasting*, *regularity*, *sharpness* — are introduced in the companion paper on distributivity and realism. That paper studies the sharpness transition (distributivity versus value-definiteness, Kochen–Specker); the paper on a σ -essential contextual state studies the regularity transition (finite versus countable additivity, resolved by a witness constructed in ZFC at ω_1); and the paper on commensurability of observation protocols studies the pasting transition (local coherence versus global realisability, with a taming catalogue and an odd-ring separating witness). We tabulate the three transitions against the coordinate in which each locates its boundary — algebraic, set-theoretic, dynamical — and record what is genuinely common: the observer-first reading, the fact that the three boundaries are different, and a method (pre-registered predictions, witness-grade verdicts, machine-checked lemmas). The two bridging claims are settled here — one negatively (the coherence notions form a strictly separated, witness-inhabited ladder rather than one notion), one positively (a compactness quarantine sorts the two constructed witnesses into provably disjoint regimes, upgrading the tabulation to a classification). A prior-art/type audit refutes the former three-way latticehood conjecture: complete Hilbert lattices fail sharpness. The surviving open theory question is the regularity predicate Φ for concrete σ -complete OMLs. This note references the three papers; it does not absorb them.

1 The observer-first reading

An observer running a protocol obtains, natively, a family of finite relative frequencies that cohere across overlapping views. That is all observation supplies. Two further properties are routinely assumed of the state such data “comes from” — that it is countably additive, and that it assigns definite truth values — but neither is given by the data. They are *representation properties*: signatures of the world being observed and of the protocol used to observe it, not of empirical adequacy itself. The question the three papers share is whether coherence, followed to its end, forces these signatures rather than assuming them, and, where it does, what exactly it forces. Their joint answer, so far partial, is a price list: what structure a world’s language and an observer’s protocol must have before countable additivity and value-definiteness come for free, and — in two cases settled by explicit construction — proofs that outside that structure the signatures can be genuinely absent while coherence remains.

This note is an umbrella. It states the shared object, tabulates how the three papers divide the labour, records the corpus-level commonalities and the genuinely corpus-level open

*Research Computing Group, Simon Fraser University, Burnaby, BC, Canada. pmahon@sfu.ca

questions, and points to each paper for the mathematics. It deliberately does not merge them: each paper travels to its own audience, and the spine carries only what no single paper can say alone.

2 The filtration and its three transitions

The companion paper on distributivity and realism introduces the organising object: on the state space of an observation algebra there is a natural filtration

$$\mathcal{S} \supseteq \mathcal{S}_\sigma \supseteq \mathcal{S}_{\text{df}}^\sigma,$$

from all coherent states, to the σ -additive ones, to the σ -additive dispersion-free ones. Here $\mathcal{S}_{\text{df}}^\sigma = \mathcal{S}_{\text{df}} \cap \mathcal{S}_\sigma$. Standalone value-definite realism is the weaker existence predicate $\mathcal{S}_{\text{df}} \neq \emptyset$; on an infinite carrier it is not silently identified with $\mathcal{S}_{\text{df}}^\sigma \neq \emptyset$. The three transitions are of different mathematical character, named there:

- (1) *pasting* — from locally coherent window-data up to a single globally realising state ($\text{EA} \rightarrow \text{PR}$);
- (2) *regularity* — from a finitely additive global state up to a countably additive one ($\text{PR}_{\text{f.a.}} \rightarrow \text{PR}_\sigma$);
- (3) *sharpness* — existence of a dispersion-free state (VDR); within the displayed filtration the endpoint also carries the separate σ -additivity requirement.

Distributivity supplies standalone sharpness and the Boolean pasting result; regularity still requires σ -additivity (and the relevant realisation condition). The corpus-level observation this note records is that **each companion paper is the deep study of exactly one transition**, and that the three locate their boundaries in different structures.

Notation fence. EA and PR are source-local symbols. In the reconstruction paper they mean overlap-compatible probability tables and realisability by one process in a declared class. In the distributivity paper they concern a dual-space measure and descent to an underlying realisation space. The architectural alignment in this note asserts no translation between those definitions.

Transition	Paper	Tame condition / status	Wild-side object
pasting	commensurability	Vorob'ev acyclicity	odd-ring half-model
regularity	σ -essential	Φ on OMLs: open	the product Ulam carrier, in ZFC
sharpness	distributivity & realism	distributivity: sufficient	Kochen–Specker (imported, reframed)

Table 1: The three transitions of the filtration, the paper that studies each, the established tame condition or boundary status, and an object witnessing the wild side. Two of the three witnesses — the odd-ring half-model and the product Ulam carrier — are constructed in the companion papers rather than imported.

The table is the weave. One question — *when does empirical adequacy purchase realism?* — is organised as a classification (made precise by the tame tower of Corollary 4.4), but not by one uniform structural boundary. Pasting has the Vorob'ev acyclicity boundary for universal extension; regularity has the open OML predicate Φ ; sharpness has distributivity as a sufficient condition and Kochen–Specker as its flagship obstruction. Two of the separating witnesses are new constructions; the third is classical and reframed here.

3 What is genuinely common

Three things hold across all three papers and belong to none of them alone.

Different structural boundaries. The commonality is architectural, not a shared dichotomy. Pasting asks for gluing and has Vorob’ev acyclicity as its universal-extension boundary. Regularity asks for finite-trace σ -liftability and leaves Φ open on concrete σ -complete OMLs. Sharpness asks for a dispersion-free state; distributivity is sufficient and Kochen–Specker blocks the complete Hilbert lattice in dimension at least three.

The observer-first price list. Section 1: the three papers are, together, the statement that σ -additivity and value-definiteness are representation properties with a computable price, and two proofs that the price is real (the two constructed witnesses).

Method. The corpus shares a working method, stated once here without ceremony: predictions are pre-registered before testing; a verdict of “realisable/tame” is admitted only from a theorem or an exact enumeration, of “contextual/wild” only from an exhibited witness; and load-bearing combinatorial lemmas are machine-checked (the σ -essential main theorem and the commensurability paper’s layer-injectivity discriminant are certified in Lean 4).

The novelty ledger, honestly. The classical mathematics each paper uses is credited there as classical (Gleason and Kochen–Specker for sharpness; Derr–Williamson, Ulam, and the Boolean baseline for regularity; Wainwright–Jordan, Vorob’ev, De Loera–Onn and stable-set theory for pasting). The corpus’s own contribution is the observer-first framing, the filtration as its organising object, and two constructed witnesses where the literature had only imported or absent ones. This is a bridge over known mathematics with two new construction points; the umbrella claims no more.

4 Two bridging results and the surviving regularity question

The two bridging claims that turn the tabulation of §2 into a single classification are settled here — one negatively (the coherence notions are a strictly separated ladder, not one notion), one positively (a compactness quarantine sorts the two constructed witnesses into disjoint regimes). The former three-way latticehood conjecture is not a third bridge: it is refuted under its literal reading, leaving one regularity question.

Proposition 4.1 (the two-gap ladder). *The coherence notions of the three papers form a strictly separated ladder,*

$$\text{pairwise-EA} \supsetneq \text{finite coherence} \supsetneq \sigma\text{-realizability},$$

and both strict inclusions are inhabited by the corpus’s own witnesses. The first gap: the golden-mean odd-ring half-model (commensurability programme, machine-verified) — the pentagon protocol C_5 over $\{0, 1\}^5$ with the half-model $s(a, b) = \frac{1}{2}$ for $a \neq b$ and 0 for $a = b$ — is pairwise-EA yet admits no finitely additive state on its generated closure: its support would have to be a proper 2-colouring of an odd cycle, of which there are none (equivalently, the stable-set bound $\sum_j \mu(a_j) \leq \alpha(C_5) = 2$ is exceeded by the half-model’s value $5/2$). The second gap: the product Ulam carrier’s pattern is finitely coherent with no σ -additive extension. Hence the three papers’ notions are not one notion evaluated at different rungs; they are the three rungs, provably distinct.

Remark 4.2. The correct corpus-level claim is therefore the rhyme, now made exact: each paper studies coherence-at-a-rung, the rungs are strictly ordered, and each strict inclusion is

witnessed inside the corpus. What survives of the “one notion” intuition is structural, not extensional: every rung is a finite-additive gluing condition over a widening class of admissible families.

Theorem 4.3 (quarantine). *Let X be a compact zero-dimensional metrizable space and $\text{Clo}(X)$ its clopen algebra.*

- (i) *Every finitely additive two-valued state s on any subalgebra $B \subseteq \text{Clo}(X)$ is the restriction of a point evaluation. Consequently a finitely coherent two-valued pattern over clopen events always has a realizing point: clauses (0) and (i) of the σ -essential recipe cannot coexist, and the σ -essential phenomenon is impossible over compact clopen carriers.*
- (ii) *Every finitely additive probability charge on $\text{Clo}(X)$ is a premeasure and extends uniquely to a Borel probability measure on X .*
- (iii) *Every charge on a subalgebra of $\text{Clo}(X)$ — in particular on the algebra generated by any protocol’s contexts — extends to $\text{Clo}(X)$, hence realizes as a Borel measure. The only obstruction to realization in the dynamical regime is finite-additive.*

Proof. (i) $F = \{C \in B : s(C) = 1\}$ is a filter of B (if $s(C) = s(D) = 1$ then $s(C \cap D) + s(C \cup D) = s(C) + s(D) = 2$ forces $s(C \cap D) = 1$, and s is monotone with $s(\emptyset) = 0$); extend F to an ultrafilter U of $\text{Clo}(X)$. U is a family of nonempty closed subsets of a compact space with the finite intersection property, so some $x \in \bigcap U$; then $s(C) = 1 \Rightarrow x \in C$ and $s(C) = 0 \Rightarrow s(C^\perp) = 1 \Rightarrow x \notin C$, i.e. $s = \delta_x \upharpoonright B$. (ii) If clopen $C = \bigsqcup_n C_n$ with all C_n clopen, compactness of C and disjointness of the open cover force all but finitely many C_n empty, so σ -additivity on $\text{Clo}(X)$ reduces to finite additivity; the Carathéodory–Hahn extension then gives a Borel measure on $\sigma(\text{Clo}(X))$, which is the Borel σ -algebra since the clopen sets form a basis, uniquely (the measure is finite, hence σ -finite; π – λ). (iii) Charge extension from a subalgebra (Horn–Tarski [2], in the finitely additive category; their countably additive analogue can fail, which is why part (ii) supplies σ -additivity separately), then (ii); see Rao–Rao [1] for the charge/Stone-space correspondence. $\square$

Corollary 4.4 (the tame tower; the witnesses are sorted). *The tame results of the corpus form a tower: finite carriers (Wright; the Boolean baseline) $\subseteq$ compact clopen carriers (Theorem 4.3) $\subseteq$ Polish-representable carriers (Derr–Williamson). In particular the two constructed witnesses inhabit provably disjoint regimes — the pasting witness lives on a compact carrier where the regularity transition is free, and the regularity witness lives at ω_1 , past the tower — so the corpus’s table is a classification, not a collection.*

Remark 4.5 (why the witness escapes; scope; credit). Theorem 4.3(i) is the Boolean baseline’s atom argument with “finite” upgraded to “compact”: the same proof at two scales, and the compactness hypothesis is exactly what the σ -essential witness escapes by living at ω_1 , where the clopen family loses the finite intersection property. The quarantine and the regularity witness are therefore consistent, not in tension: the theorem proves the witness *cannot* descend to the compact dynamical regime. The theorem quarantines realization by *measures*; whether exotic σ -classes generated inside a Borel structure satisfy the lifting property Φ is the σ -essential paper’s open characterization question, untouched here. Parts (ii)–(iii) are Stone duality for charges [2, 1], classical; the corpus’s contribution is part (i)’s pointed form, the tower, and the sorting.

The literal three-way conjecture fails a type check and a counterexample check. The corpus defines liftability exactly only at regularity, through the finite-trace predicate Φ ; and $L(H)$ for $\dim H \geq 3$ is already a complete orthomodular lattice while Kochen–Specker gives $\mathcal{S}_{\text{df}}(L(H)) = \emptyset$. Latticehood therefore cannot imply sharpness.

The surviving theory-front question is: does every concrete, σ -complete, non-Boolean, essentially irreducible orthomodular lattice satisfy Φ ? Equivalently, whenever a two-valued state on a finite orthocomplement-closed substructure is the trace of a global finitely additive two-valued state, is the same finite trace realised by some global σ -additive two-valued state? This question belongs wholly to the regularity transition. The pasting and sharpness rows retain their own boundaries and are not instances of it.

5 Reading and dissemination order

For a reader: distributivity & realism first (it introduces the filtration), then σ -essential (the deepest single result), then commensurability (the broadest). For dissemination the crown result travels first and alone: the σ -essential paper is self-contained and Lean-certified, and its Derr–Williamson cell-closure framing does not depend on the other two. The commensurability paper follows; the distributivity paper may belong to a different venue class (foundations of physics) than the two mathematics papers. The spine lets each paper travel light while carrying, itself, only the corpus-level content: the filtration as shared object, the observer-first moral, the two bridging results, and the surviving regularity question of §4.

References

- [1] K. P. S. Bhaskara Rao and M. Bhaskara Rao. *Theory of Charges: A Study of Finitely Additive Measures*. Academic Press, London, 1983.
- [2] Alfred Horn and Alfred Tarski. Measures in Boolean algebras. *Transactions of the American Mathematical Society*, 64(3):467–497, 1948.